Performance engineering of semiconductor spin qubit systems

Ramin M. Abolfath[1,2], Thomas Brabec[1]
[1] *University of Ottawa, Physics Department 150 Louis Pasteur, Ottawa, ON, K1N 6N5, Canada*
[2] *School of Natural Sciences and Mathematics, University of Texas at Dallas, Richardson, TX 75080*
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The performance of a quantum computation system is investigated, with qubits represented by magnetic impurities in coupled quantum dots filled with two electrons. Magnetic impurities are electrically manipulated by electrons. The dominant noise source is the electron mediated indirect coupling between magnetic impurities and host spin bath. As a result of the electron mediated coupling, both noise properties and the time needed for elementary gate operations, depend on controllable system parameters, such as size and geometry of the quantum dot, and external electric and magnetic fields. We find that the maximum number of quantum operations per coherence time for magnetic impurities increases as electron spin singlet triplet energy gap decreases. The advantage of magnetic impurities over electrons for weak coupling and large magnetic fields will be illustrated.

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Semiconductor nanostructures [1] are a promising host material for the realization of quantum computers [2] because of the well developed production technologies and because of the potential for scaling to multi qubit systems. The greatest bottleneck for realizing semiconductor quantum devices is the high degree of noise present in the host material.

Recently substantial progress has been made in the experimental demonstration of electron spin coded qubits in lateral double QDs [3], first theoretically proposed by Loss and DiVincenzo [4]. This system exhibits promising properties for quantum information processing, as it can be controlled electrically like charge qubits, however offers the longer decoherence times of spin qubits. Nevertheless, dephasing in spin coded qubits due to nuclear hyperfine interaction still presents the major hurdle for realizing a scalable quantum computer [5, 6].

Another seminal idea of realizing quantum computation in semiconductors is by the use of a hybrid qubit, first proposed by Kane in a ^{31}P doped silicon host [2, 7, 8]. In silicon, ^{31}P is a positive dopant that is transformed into a positively charged nucleus and a loosely bound valence electron. Whereas, the qubit is represented by the nuclear spin of ^{31}P, electric gate operations on the electron are used for manipulation and diagnostics of the qubit. The technological implementation of this system is more challenging, mostly as a result of the significantly smaller dimensions of the qubits.

We suggest and analyze here a qubit that is a mixture of the spin coded [4] and of the hybrid qubits [7, 8] discussed above. The proposed device consists of a QD containing electrons and a neutral dopant acting as a magnetic impurity (MI). Similar to the original hybrid qubit, the MI represents the qubit, and the electrons are used for manipulating the qubit. The use of MIs alone for spin-based quantum devices and its noise spectroscopy has been investigated recently [9, 10, 12]. Here, we confine our analysis to MIs with zero nuclear spin, [e.g., ^{56}Fe/^{160}Gd doped in II-VI/IV-VI materials], to exclude coupling between spins of the electron and nucleus of the MI. As shown in Fig.1, the MI is localized in space and its direct interaction with the host spin bath is negligible. Therefore, the decoherence of the qubit is determined by the electron mediated coupling of the MI to the spin bath.

Our configuration has two advantages over the original qubit systems. First, it allows greater design flexibility, in particular with regard to the size of the confining potential. This will facilitate the actual technological realization of a hybrid qubit. Second, MI-electron coupling and electron mediated MI-spin bath coupling show particular dependence on the electron spin singlet triplet energy gap $\Delta_{\rm e}$, that allows forming stable qubit over a range of QD confining potentials, and the external electric and magnetic fields. This system corresponds to the electron spin coded qubit system investigated in Ref. [4] and makes a comparison possible. The performance of our quantum computing setup is measured by the maximum number of operations $N_i = \tau_i/T_i$ with $i = e, m$ for electrons and MI qubits, respectively. Here, τ is the coherence time and T is the time required for an elementary gate operation. In this work we illustrate that the quantum performance of the spin coded qubit based on MIs increases by the external electric and magnetic fields. This is in contrast to the electron spin coded qubit that its performance is suppressed rapidly by the inter-dot coupling.

We represent the QD system by the Hamiltonian $H = H_{\rm e} + H_{\rm m} + H_{\rm n} + H_{\rm in}$, which contains electrons ($H_{\rm e}$), MIs ($H_{\rm m}$), host semiconductor nuclei ($H_{\rm n}$), and their interactions ($H_{\rm in}$). In our model we consider e-MI and e-n interaction, i.e. $H_{\rm in} = H_{\rm em} + H_{\rm en}$. The direct interaction between MI electrons and semiconductor host nuclei are neglected, as d- and f-electrons are highly localized in space. Further, our analysis is confined to MIs with zero nuclear magnetic moment, which do not interact with the MI electrons. As a result, dipole-dipole interaction